THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · HARD LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A pea plant RrYy is crossed with rryy. Out of 200 offspring, the number of round green plants is expected to be 1

(a) 25 **(b)** 100 **(c)** 50 **(d)** 150

2. A concave mirror forms a real image of a candle on a screen kept at 45 cm. If the candle is 15 cm from the mirror, the focal length of the mirror (magnitude) is 1

(a) 15 cm **(b)** 11.25 cm **(c)** 30 cm **(d)** 22.5 cm

3. A person with a far point at 25 cm requires spectacles to see distant objects clearly. The focal length of the lens is 1

(a) +25 cm **(b)** -31 cm **(c)** -6 m **(d)** -25 cm

4. The resistance of a wire of length L and area A is 12 ohm. A second wire of the same material has length 2L and radius double that of the first. Its resistance is 1

(a) 12 ohm **(b)** 3 ohm **(c)** 6 ohm **(d)** 24 ohm

5. A straight wire with current is placed between the poles of a magnet. The force on the wire is greatest when the wire makes an angle with the field of 1

(a) 0 degrees **(b)** 90 degrees **(c)** 45 degrees **(d)** 30 degrees

6. A food chain has four trophic levels. If the energy at the fourth level is 5 J, the energy at the first level is 1

(a) 5000 J **(b)** 500 J **(c)** 50,000 J **(d)** 50 J

7. A metal X turns black on heating in air. The black product, on passing hydrogen over it, turns reddish brown. X is: 1

(a) iron **(b)** copper **(c)** zinc **(d)** magnesium

8. A white powder X on adding water forms a hard solid mass. X is: 1

(a) plaster of Paris **(b)** baking soda **(c)** washing soda **(d)** bleaching powder

9. A student heats a green carbonate ore of copper. The black residue and gas are: 1

(a) Cu_2O and O_2 **(b)** Cu and CO_2

(c) CuO and CO **(d)** CuO and CO_2

10. A hydrocarbon X decolourises bromine water and has a molecular mass of 28 u. X is: 1

(a) ethane **(b)** ethyne **(c)** methane **(d)** ethene

11. A closed jar contains a green plant in sunlight together with a mouse. The mouse survives longer than a mouse alone in a jar because the plant: 1

(a) uses carbon dioxide and releases oxygen **(b)** releases carbon dioxide
(c) gives out nitrogen **(d)** absorbs oxygen only

12. A boy has a rapid pulse, sweating and wide pupils when frightened. This is due to: 1

(a) insulin **(b)** thyroxine **(c)** adrenaline **(d)** oestrogen

13. An organism with a very complex body that can still regenerate would need 1

(a) no cells at all
(b) only gametes
(c) no DNA
(d) undifferentiated cells to produce missing parts

14. A pea plant that is tall and round (TtRr) is crossed with dwarf wrinkled (ttrr). The offspring ratio is 1

(a) 3 : 1 **(b)** 9 : 3 : 3 : 1 **(c)** 1 : 1 **(d)** 1 : 1 : 1 : 1

15. An object is placed 24 cm from a lens and a real image forms at 24 cm. The lens is 1

(a) concave, f = -24 cm **(b)** convex, f = 12 cm
(c) convex, f = 24 cm **(d)** concave, f = -12 cm

16. The sky is blue because 1

(a) the sun emits blue only
(b) smaller wavelengths are scattered more by small particles
(c) the sea reflects blue
(d) larger wavelengths are scattered more by small particles

17. **Assertion (A):** Guard cells control the opening and closing of stomata. 1
Reason (R): Guard cells swell when water flows into them, causing the stomatal pore to open.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** A reflex action is a quick, automatic response. 1
Reason (R): In a reflex action the signal passes through the spinal cord and not through the brain.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

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19. **Assertion (A):** Plants raised by vegetative propagation show characters identical to the parent plant. 1

Reason (R): Vegetative propagation involves only one parent and no fusion of gametes.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. **Assertion (A):** When pure tall (TT) and pure dwarf (tt) pea plants are crossed, all the F1 plants are tall. 1

Reason (R): Tallness is the dominant trait over dwarfness.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. A tiny part of a Hydra body is used to grow a new Hydra in a lab. Is this regeneration or budding? Justify. 2

22. In a cross of two heterozygous tall plants (Tt x Tt), 240 plants were produced. Find the number of tall plants that are homozygous. 2

23. The image of an object formed by a concave mirror is real, inverted and 1/3 of the object size at 15 cm from the mirror. Find the object distance and focal length. 2

24. Why does the amount of scattering of red light not matter much in the colour of the sky? 2

25. Calculate the number of electrons flowing per second in a wire carrying 4.8 mA. 2

26. A charged particle enters a magnetic field and its speed remains unchanged. Explain why. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. A concave mirror forms a real image three times larger than the object. When the object is moved 10 cm farther away, the image becomes the same size as the object. Find f. 3

28. A person with a near point at 30 cm wants to read at 25 cm. Find the power of the lens required. 3

29. The resistance of a copper wire is 2.5 ohm at 20 degrees Celsius. Why does its resistance rise at high temperature, and what is the effect on the current if the voltage is constant? 3

30. Explain why the force on a current-carrying conductor depends on the angle between the conductor and the field. 3

31. In a pond food chain with plankton, small fish, big fish and kingfisher, the concentration of a pesticide in the plankton is 0.02 ppm. If it rises 10 times at each level, find the concentration in the kingfisher. 3

OR

Green coating forms on copper vessels kept in humid air. Name the coating, give its cause and one prevention. 3

32. A white solid X on heating gives CO_2 and a residue Y. Y dissolves in water, is basic and is used in whitewash. Identify X and Y and write equations. 3

OR

Why does iron rust faster near the sea? Explain the role of salt in rusting. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** In an experiment, equal-sized pieces of magnesium, zinc, iron and copper were dropped into separate test tubes containing dilute hydrochloric acid. Bubbles of a gas appeared vigorously on magnesium, slowly on zinc, very slowly on iron and no bubbles appeared on copper. 5

(i) Which metal does not react with dilute HCl?

(ii) Name the gas evolved and give its test.

(iii) Write the equation for the reaction of zinc with dilute HCl.

(iv) Arrange the four metals in decreasing order of reactivity and state the conclusion drawn from the rate of bubbling.

34. **Case study.** Alkanes form a homologous series with the general formula C_nH_{2n+2}. The first member is methane, the second is ethane and the third is propane. The fourth member is butane and the fifth is pentane. 5

(i) Write the molecular formula of the alkane with five carbon atoms.

(ii) Name the first member of the alkyne series.

(iii) What is the difference in molecular mass between two consecutive members of a homologous series?

(iv) Find the number of covalent bonds in propane and explain why propane is called a saturated compound.

35. A straight wire carries current from west to east in a field pointing north. Find the direction of the force and describe how it changes 5

(a) when the current is reversed

(b) when both are reversed. State the rule used.

36. In a food chain: Phytoplankton, Zooplankton, Small fish, Large fish, Fish-eating birds, the energy at the producer level is 2,000,000 J. Calculate the energy at each level using the 10 per cent law. At which level are the largest losses in terms of the actual energy? 5

OR

Ferrous sulphate crystals are heated. Describe the colour changes, write the equation, name the gases and the type of reaction. 5

37. Write a note on water of crystallisation, giving three examples with colour change on heating and one test. 5

OR

Explain: 5

(a) why sodium is kept in kerosene

(b) why aluminium foil is used in packaging

(c) why gold is alloyed

(d) why ionic compounds are brittle.

38. Draw structures and name all isomers of C_4H_{10}, C_4H_8 positions (but-1-ene, but-2-ene) and explain the difference between structural isomers and homologues. 5

OR

Compare ATP production in aerobic respiration, anaerobic in yeast and anaerobic in muscle, with the sites and products and the reason for the energy difference. 5

END OF QUESTION PAPER **Total: 80 Marks**

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